

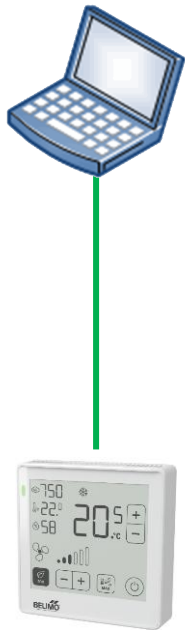
B2 - BACnet / Modbus (Advanced) Modbus - Workshop I

ROU

B2 - Modbus – Workshop I

ROU- Network

Modbus Poll



P-22RTM-1U00D-2

— Serial network (RS485)

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1. Power your device.

 *See wiring diagrams in the datasheet*

2. Connect your device via Modbus RTU with your laptop.

 *Use RS485-USB adapter*

 *See wiring diagram in the datasheet*

3. Configure your room operating unit according to network overview.

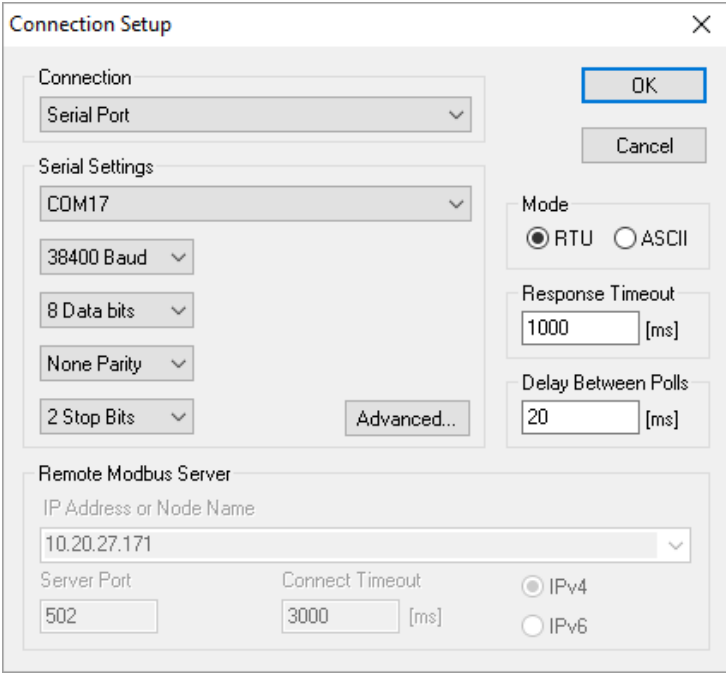
 *Find your appropriate configuration tool in the documents (LINK.10, Belimo Assistant 2 etc.)*

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4. Start Modbus Poll. Open a *Serial Port Connection*.

- 💡 Check on which COM port your RS485-USB adapter is
- 💡 Pay attention to the Modbus settings (on the device and in the bus tool)



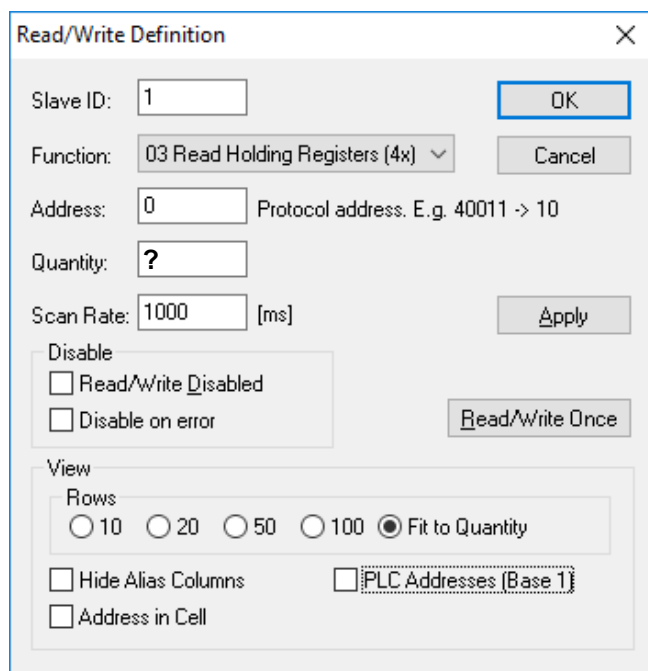
The image shows a 'Connection Setup' dialog box with the following fields and options:

- Connection:** A dropdown menu set to 'Serial Port'.
- Serial Settings:**
 - COM17:** A dropdown menu.
 - 38400 Baud:** A dropdown menu.
 - 8 Data bits:** A dropdown menu.
 - None Parity:** A dropdown menu.
 - 2 Stop Bits:** A dropdown menu.
 - Advanced...** A button.
- Mode:** Radio buttons for **RTU** (selected) and **ASCII**.
- Response Timeout:** A text box containing '1000' with '[ms]' next to it.
- Delay Between Polls:** A text box containing '20' with '[ms]' next to it.
- Remote Modbus Server:**
 - IP Address or Node Name:** A text box containing '10.20.27.171'.
 - Server Port:** A text box containing '502'.
 - Connect Timeout:** A text box containing '3000' with '[ms]' next to it.
 - IPv4:** A selected radio button.
 - IPv6:** An unselected radio button.
- Buttons:** 'OK' and 'Cancel' buttons.

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5. Open a new window  and press the F8 key.
Make the settings according to the picture on the left.
Repeat the procedure with the settings as shown in the other pictures.



Read/Write Definition

Slave ID: 1 OK

Function: 03 Read Holding Registers (4x) Cancel

Address: 0 Protocol address. E.g. 40011 -> 10

Quantity: ?

Scan Rate: 1000 [ms] Apply

Disable

☐ Read/Write Disabled

☐ Disable on error Read/Write Once

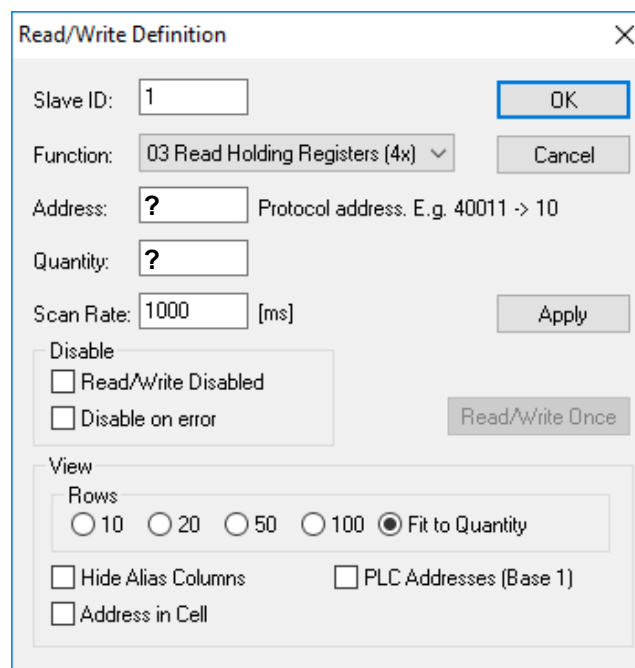
View

Rows

☐ 10 ☐ 20 ☐ 50 ☐ 100 ☒ Fit to Quantity

☐ Hide Alias Columns ☐ PLC Addresses (Base 1)

☐ Address in Cell



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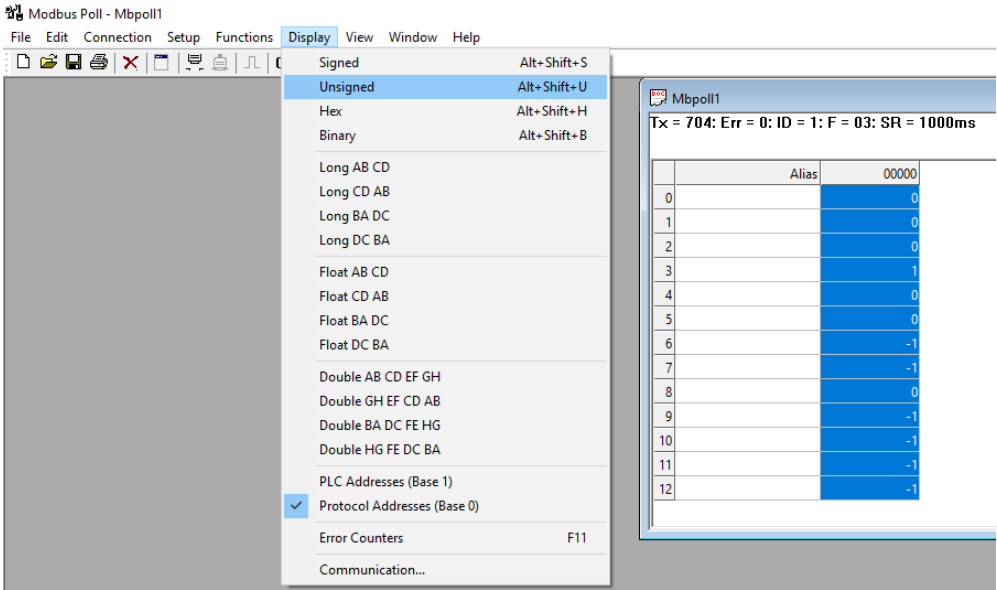
☐ Hide Alias Columns ☐ PLC Addresses (Base 1)

☐ Address in Cell

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6. Display all values as *Unsigned Integer*.



7. What are the main Modbus blocs/section?
e.g. in register No. 1 – 6 you find sensor values

1. <i>Sensor values</i>	2.	3.
4.	5.	6.
7.	8.	9.
10.	11.	

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9. Read out the current room temperature, relative humidity and CO2

1. Temp:	Interpretation:
2. Rh:	Interpretation:
3. CO2:	Interpretation:

7. Set [ModeOnOffButton] to “Selection”

Set [EnableLocalAdjustment] to “Enabled”

Check if On-Off button is visible on ROU display Yes ☐ No ☐

Check if any adjustment is possible on ROU display Yes ☐ No ☐

7. Set [DisplayTemperature] , [DisplayRelHumidity] and [DisplayCO2] to “Visible”

Check if actual temperature value is shown on ROU display Yes ☐ No ☐

Check if actual rel. humidity value is shown on ROU display Yes ☐ No ☐

Check if actual CO2 value is shown on ROU display Yes ☐ No ☐

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9. Set the [Display unit] to ° F.

Check if temperature sensor values shown on Modbus Poll and ROU display are different because of different temp. units. ☐ Yes ☐ No

ActualTemp@Modbus Poll:

ActualTemp@Display:

Set [Display unit] back to ° C

Check if display and Modbus Poll show same values. ☐ Yes ☐ No

ActualTemp@Modbus Poll:

ActualTemp@Display:

10. Set [OffsetTemperature] to 2° C. What is the new actual temp. value?

ActualTemp:

Interpretation:

Set [OffsetTemperature] back to 0° C as default.

2. ActualTemp:

Interpretation:

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11. Read serial number.

Values:

Interpretation:

12. Change the color of the display background. Which register No. did you use?
